

Chapter 1, Problem 28P

Problem

A 150-W incandescent outdoor lamp is connected to a 120-V source and is left burning continuously for an average of 12 hours per day. Determine:

(a) the current through the lamp when it is lit.

(b) the cost of operating the light for one non-leap year if electricity costs 9.5 cents per kWh.

Step-by-step solution

Step 1 of 2

(a)

Consider the following data:

The power rating of the bulb is, $p = 150 \text{ W}$

The voltage rating of the bulb is, $v = 120 \text{ V}$

The average running time of bulb is, $t = 12 \text{ h}$

Calculate the current through the lamp.

$$\begin{aligned} i &= \frac{p}{v} \\ &= \frac{150 \text{ W}}{120 \text{ V}} \\ &= 1.25 \text{ A} \end{aligned}$$

Therefore, the current through the bulb when it is lit is $\boxed{1.25 \text{ A}}$.

Step 2 of 2

(b)

Calculate the energy consumed by the bulb for one non-leap year.

$$\begin{aligned} w &= pt \\ &= (150 \text{ W})(365)(12 \text{ h}) \\ &= 657 \text{ kWh} \end{aligned}$$

Electricity costs 9.5 cents per kWh.

Calculate the total cost of operating the bulb.

$$\begin{aligned} \text{cost} &= (657 \text{ kWh})(9.5 \text{ cents/kWh}) \\ &= 6241.5 \text{ cents} \end{aligned}$$

Therefore, the total cost of operating the bulb for one non-leap year is $\boxed{6241.5 \text{ cents}}$.